

Automodification, Timing, and the Problem of Unnecessary Intervention

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ABSTRACT

This paper develops an epistemic and dynamical-systems interpretation of a recurring observation: "Don't you think it's strange they always show up on time when unnecessary but never explain its worth?"

The statement can be formalized without assuming that an unexplained timing pattern proves intentional coordination. The central research problem is instead whether the perceived relationship between timing, intervention, necessity, and explanatory value can be measured.

The proposed framework treats an intervention as an event I_i occurring at time t_i , with an estimated necessity N_i , observable effect E_i , and stated or inferred value V_i . A potentially informative quantity is the discrepancy between the intervention's temporal predictability and its demonstrated utility:

$$\Delta_i = P(I_i \mid t_i, \mathcal{H}) - U(I_i)$$

where $\mathcal{H}$ denotes the available history and U denotes independently measured utility.

The paper introduces an Automodification Hypothesis: systems that respond to their own outputs, expectations, or monitoring criteria can produce intervention patterns that appear strangely well-timed even when no external coordinator is assumed. In biological systems, genuine automodification is a recognized phenomenon, and mathematical models show that self-modification can produce unintuitive system-level behavior. Similar concepts can be generalized cautiously to information-processing systems, institutions, algorithms, and human decision processes. [1–3]

The objective is therefore not to establish that unexplained timing demonstrates surveillance, manipulation, or intentional coordination. Rather, it is to construct a falsifiable method for

distinguishing coincidence, feedback, selection effects, anticipation, ordinary scheduling, and genuine adaptive coupling.

1. Introduction

Human reasoning is highly sensitive to temporal coincidence. When an event repeatedly occurs near a moment at which it becomes noticeable, consequential, or unnecessary, an observer may infer a causal relationship. The inference can be compelling even when the underlying mechanism is unknown.

Consider the qualitative pattern:

event appears → event is temporally appropriate → event is nevertheless un:



This produces an epistemic asymmetry. The timing appears informative, while the purpose remains obscure. The resulting question is:

Why does the intervention appear temporally optimized if its utility is unexplained?

There are many possible answers. The event may be:

1. genuinely coordinated;
2. generated by an ordinary schedule;
3. triggered by a feedback mechanism;
4. selected because only salient coincidences are remembered;
5. predicted by a system responding to observable conditions;
6. caused by an unrecognized common variable;
7. genuinely random but retrospectively interpreted as meaningful.

The scientific problem is to distinguish these explanations.

2. From Suspicion to Measurement

A useful formalization begins by avoiding assumptions about who or what caused an event. Define an intervention:

$$I_i = (t_i, x_i, a_i)$$

where:

t_i = time of occurrence,

x_i = observable context,

a_i = observable action or intervention.

For each intervention define:

$$N_i = \text{necessity}, \quad U_i = \text{measured utility}, \quad V_i = \text{explained value}$$

The intuitive observation can then be expressed as:

$$N_i \rightarrow 0, \quad U_i \rightarrow 0, \quad V_i \rightarrow 0$$

while temporal predictability remains high:

$$P(I_i \mid x_i, \mathcal{H}) \rightarrow 1.$$

This is the interesting regime. It does not establish intentionality. It identifies a statistical phenomenon worthy of investigation.

3. The Timing–Utility Paradox

Suppose a sequence of interventions is highly temporally predictable but contributes little measurable utility. Define temporal predictability:

$$T_p = I(I; t \mid \mathcal{H})$$

and intervention utility:

$$U = \mathbb{E}[\Delta Q \mid I]$$

where $I(\cdot; \cdot)$ is mutual information and Q is the quality of the relevant outcome.

A high- T_p , low- U regime produces:

$$T_p \gg 0, \quad U \approx 0$$

This is the formal version of the intuition that something is "always on time" while failing to explain why its presence matters. Several mechanisms can generate this combination.

4. Automodification

The term automodification has a precise meaning in molecular biology: a biological component modifies itself or changes its own functional state. PARP enzymes, for example, undergo self-ADP-ribosylation, and mathematical models of biochemical automodification can produce counterintuitive system-level behavior. [1–3]

This concept provides a useful analogy for information systems. A generalized adaptive system can be represented as:

$$S_{t+1} = F(S_t, I_t)$$

while the intervention policy itself changes according to the system's history:

$$\pi_{t+1} = G(\pi_t, S_t, I_t).$$

The second equation is the important addition. The system is not merely acting on its environment. It is modifying the policy that determines how it acts. This produces a feedback loop:

observation $\rightarrow$ intervention $\rightarrow$ system modification $\rightarrow$ new observation $\rightarrow$ n



In such systems, apparently mysterious timing can arise naturally from feedback.

5. The "Always on Time" Effect

Consider an observer who expects an event around time t^* . An adaptive system may respond to the same conditions that generated the observer's expectation:

$$X(t) \rightarrow I(t)$$

while the observer independently estimates:

$$X(t) \rightarrow \hat{I}(t).$$

The two predictions can become correlated without requiring direct access to the observer's thoughts. Thus:

$$P(I | X) \gg P(I)$$

can produce an apparent coincidence. The critical distinction is:

correlation with observable conditions $\neq$ access to private cognition

This distinction prevents an ordinary feedback process from being mistaken for a much stronger claim.

6. Why Unnecessary Events Can Still Appear

An intervention can be unnecessary from the observer's perspective while remaining optimal according to another objective function. Suppose a system optimizes:

$$J = \alpha U + \beta R + \gamma C$$

where:

U = direct usefulness,

R = risk reduction,

C = cost.

The observer may evaluate only U . The system may optimize all three. An intervention with:

$$U \approx 0$$

can nevertheless be rational if:

$$\beta R - \gamma C > 0.$$

This provides an important alternative explanation for the phrase "never explain its worth." The worth may exist, but it may be encoded in a variable that the observer does not measure.

7. Explanatory Value as an Information Variable

A stronger definition of "worth" is informational. Let H represent uncertainty about the system's future state. An intervention has informational value if:

$$\Delta H = H(S_t) - H(S_t | I_t) > 0.$$

Thus an intervention can be useful even if it produces no obvious external benefit. It may:

- reduce uncertainty,
- test a hypothesis,
- maintain system stability,
- calibrate a sensor,
- preserve a fallback state,
- detect anomalies,
- prevent a low-probability failure.

Consequently:

visible utility $\neq$ total system utility.

This is why the correct experiment must measure consequences rather than infer purpose from appearance.

8. The Temporal Selection Problem

Human memory introduces another mechanism. Suppose events occur randomly with rate λ . The probability of at least one event occurring within a window Δt is:

$$P(N \geq 1) = 1 - e^{-\lambda \Delta t}.$$

If the observer remembers only unusually well-timed events, the remembered sample is not representative of the complete event distribution. Let:

$$M_i = \begin{cases} 1, & \text{event is remembered} \\ 0, & \text{otherwise.} \end{cases}$$

If:

$$P(M = 1 \mid \text{salient timing}) > P(M = 1 \mid \text{ordinary timing}),$$

then the remembered dataset becomes biased toward coincidences. This creates a selection mechanism that can make random events appear non-random.

9. A Bayesian Model

The appropriate epistemic question is not:

$$P(\text{coordination} \mid \text{coincidence}) = 1.$$

Instead compare competing hypotheses.

H_0 : Coincidence

$$I_t \sim P_0(I \mid X_t)$$

H_1 : Adaptive feedback

$$I_t \sim P_1(I \mid X_t, \mathcal{H}_t)$$

H_2 : External coordination

$$I_t \sim P_2(I \mid X_t, Z_t)$$

where Z_t represents an additional coordinating variable. The evidence should update:

$$P(H_k \mid D) \propto P(D \mid H_k) P(H_k).$$

Repeated coincidence alone is therefore insufficient. What matters is whether the observations have a substantially higher likelihood under one model than its competitors.

10. A Controlled Experimental Design

A rigorous experiment would record all relevant events rather than only memorable ones. The central metric could be:

$$A_i = P_i - \hat{U}_i$$

Variable	Definition
t_i	Event time
x_i	Observable context
a_i	Intervention
N_i	Estimated necessity
U_i	Measured utility
V_i	Stated rationale
R_i	Outcome/risk change
C_i	Cost
P_i	Pre-event predicted probability

where A_i represents the mismatch between expected timing and demonstrated utility. Across many events:

$$\bar{A} = \frac{1}{n} \sum_i A_i.$$

A consistently large $\bar{A}$ would justify further investigation. It would not identify the mechanism by itself.

11. Intervention Efficiency

Define intervention efficiency as:

$$\eta_i = \frac{U_i}{C_i + \epsilon}.$$

Then compare efficiency with temporal predictability. Three regimes become particularly interesting:

Efficient and predictable

$$T_p \gg 0, \quad \eta \gg 0.$$

Ordinary optimization may explain the observations.

Inefficient and unpredictable

$$T_p \approx 0, \quad \eta \approx 0.$$

Random or poorly optimized events may explain the observations.

Predictable but inefficient

$$T_p \gg 0, \quad \eta \approx 0$$

This is the regime most closely corresponding to the original intuition. It deserves investigation because conventional utility optimization does not immediately explain it.

12. Automodification as a Candidate Mechanism

An adaptive system may generate the third regime if its internal objective is not the same as the observer's objective. Let:

$$\pi_t = \arg \max_{\pi} J_t(\pi).$$

If the system modifies its own objective:

$$J_{t+1} = G(J_t, S_t),$$

then the intervention policy becomes self-referential:

$$\pi_{t+1} = F(\pi_t, S_t).$$

This can produce apparently redundant actions because the system is maintaining or modifying the conditions under which future decisions are made.

In biological systems, self-modification can generate unintuitive responses because modification changes the availability or effectiveness of the very component performing the modification. Mathematical modeling of SUMO and PARP systems provides concrete examples of this general principle. [1–3]

The analogy should be used cautiously: biological automodification is experimentally established, while applying the concept to social or computational observations

requires independent evidence.

13. The Value-Explanation Gap

The phrase "never explain its worth" identifies a second variable that is often neglected. Define:

$$G_V = V_{\text{required}} - V_{\text{explained}}.$$

If:

$$G_V \gg 0,$$

then the event has a large value-explanation gap. The combination

$$T_p \gg 0, \quad \eta \approx 0, \quad G_V \gg 0$$

defines an Unexplained Timely Intervention (UTI) pattern. This is proposed here as a descriptive category, not as evidence of any particular cause.

14. Falsification

The hypothesis should be rejected if ordinary variables explain the timing. Potential confounds include:

- fixed schedules;
- known deadlines;
- environmental cycles;
- notification systems;
- habitual behavior;

- shared information;
- predictable workload;
- attention and memory biases;
- regression to the mean;
- selective recording.

The strongest test is therefore preregistration. Before observing the next sequence:

$$\mathcal{H}, \quad P(I), \quad \Delta t, \quad U, \quad C$$

should be specified in advance. Post-hoc pattern discovery should be treated as hypothesis generation rather than confirmation.

15. An Information-Theoretic Interpretation

Suppose an intervention provides no measurable utility but carries substantial information about the state of a system. Then:

$$U \approx 0$$

while

$$I(I; S) > 0.$$

This produces an important distinction:

instrumental value $\neq$ informational value

An apparently unnecessary intervention may therefore be a measurement, calibration event, probe, or feedback operation. The observer may interpret it as purposeless

because the information being generated is not visible.

16. Relation to Consciousness and the Spatiotemporal Hypothesis

This framework connects naturally with the preceding photonic and spatiotemporal research program. A biological observer is itself an adaptive dynamical system:

$$E_t \rightarrow R_t \rightarrow B_t \rightarrow A_t \rightarrow E_{t+1}.$$

The observer continuously predicts events and updates internal models. Thus the perception of an event being "on time" may itself be generated by the same temporal predictive machinery under investigation. The deeper question becomes:

Is the observed timing an external property, an internal prediction, or a coupled property of both?

This can be experimentally investigated by comparing:

$$T_{\text{external}}, \quad T_{\text{neural}}, \quad T_{\text{reported}}.$$

If these temporal variables systematically synchronize, the apparent coincidence may emerge from coupling rather than from a mysterious independent mechanism.

17. Toward an Automodifying Observer

A more exotic theoretical model treats the observer as a system that modifies the very variables by which it predicts the world. Let the internal model be:

$$M_{t+1} = F(M_t, D_t)$$

where D_t is incoming data. But the data distribution itself may depend on the observer's actions:

$$D_{t+1} = G(M_t, A_t, E_t).$$

The combined system is therefore:

$$(M_t, E_t) \rightarrow A_t \rightarrow (M_{t+1}, E_{t+1})$$

This is a closed adaptive loop. In such a system, the observer does not merely perceive temporal structure. It participates in producing the future distribution from which temporal structure is inferred.

18. Discussion

The original intuition — "they always show up on time when unnecessary but never explain its worth" — contains two distinct empirical claims:

1. a temporal claim;
2. an explanatory-value claim.

They should not be conflated. The temporal claim can be measured statistically. The value claim can be evaluated through outcomes and stated objectives. The relationship between them can then be tested. A striking pattern would be:

high temporal predictability + low demonstrated utility + high unexplained



Such a pattern would justify searching for hidden variables, feedback mechanisms, selection effects, or alternative objective functions. It would not by itself establish an intentional actor. This distinction is central to maintaining scientific validity.

19. Proposed Research Program

A multi-stage research program follows naturally.

Phase I — Event logging. Record every relevant event, including events that do not appear meaningful.

Phase II — Temporal analysis. Estimate $P(I_t \mid X_t, \mathcal{H}_t)$.

Phase III — Utility analysis. Measure ΔQ_t before and after intervention.

Phase IV — Explanation analysis. Record whether an independently verifiable rationale exists.

Phase V — Feedback analysis. Determine whether previous events alter subsequent intervention probabilities.

Phase VI — Automodification analysis. Test whether the intervention process itself changes over time ($\pi_{t+1} \neq \pi_t$).

Phase VII — Model comparison. Compare coincidence, ordinary scheduling, adaptive feedback, and other hypotheses using preregistered statistical criteria.

20. Conclusion

An event that repeatedly appears "on time" while appearing unnecessary and unexplained is psychologically striking, but its scientific interpretation is underdetermined. The correct response is neither immediate dismissal nor immediate attribution of hidden coordination. It is measurement. The central research object is the relationship:

$$\text{timing} \leftrightarrow \text{necessity} \leftrightarrow \text{utility} \leftrightarrow \text{explanation}$$

The concept of automodification provides an especially interesting theoretical possibility because self-modifying systems can exhibit behavior that is unintuitive

from the perspective of an external observer. Biological examples demonstrate that feedback through self-modification can generate unexpected system-level dynamics. [1–3]

The broader hypothesis is therefore:

apparent mysterious timing $\neq$ proof of external agency

but potentially:

apparent mysterious timing = f (feedback, prediction, selection, hidden object)



The scientific opportunity lies in determining which terms are actually present. An intervention should not be judged solely by whether it appears unnecessary. It should be evaluated by what changes because it occurred, what information it carries, what objective it serves, and whether its timing can be predicted from independently measurable variables.

Only then can the apparent paradox of "always on time, yet never worth explaining" be converted from an intuition into a testable theory.

References

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2. Suskiewicz, M. J. et al. (2021). Progress and outlook in studying the substrate specificities of PARPs and related proteins. *FEBS Journal*. Reviews PARP automodification and its biochemical context.
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Scientific Status: This paper is a conceptual and theoretical research proposal. The existence of biological automodification is established, but extending the concept to unexplained human or social timing patterns is a hypothesis requiring controlled evidence. No claim of surveillance, intentional coordination, or access to private mental states follows from temporal coincidence alone.

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